

No. of Printed Pages : 4

Register Number

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**PART – III**, awgpay / **PHYSICS**

(English Version)

Time Allowed : 3.00 Hours]

[Maximum Marks : 70

- Instructions :**
- (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.
 - (2) Use **Blue** or **Black** ink to write and underline and pencil to draw diagrams.

PART – I

- Note :**
- (i) Answer **all** the questions. **15x1=15**
 - (ii) Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

1. The magnitude of electric intensity at a distance 'r' from the centre of an electric dipole along its axial line is E. The distance of the point from the centre of the electric dipole along its equatorial line at which the electric intensity has same value E is :
 - (a) $\frac{r}{\sqrt{2}}$
 - (b) $\frac{r}{(3)^{\frac{1}{3}}}$
 - (c) $r(2)^{\frac{1}{3}}$
 - (d) $\frac{r}{(2)^{\frac{1}{3}}}$
2. A beam of protons and α -particle are successively accelerated in a cyclotron. The ratio of the normal magnetic field to be applied to the cyclotron so that protons and α -particles have the same period of rotation is :
 - (a) 1:4
 - (b) 4:1
 - (c) 1:2
 - (d) 2:1
3. Two sample of radioactive substances have the same quantity. $\frac{1}{16}$ th portion of A and $\frac{1}{256}$ th portion of B remain un-decayed after 8 hours. The ratio of half-life periods of A and B is :
 - (a) 1:4
 - (b) 4:1
 - (c) 1:2
 - (d) 2:1
4. The threshold frequency of a photo-sensitive surface is 5×10^{14} Hz. Then which of the following can produce photoelectric emission from the same surface?
 - (a) Ruby-Laser light
 - (b) He-Ne Laser light
 - (c) Xenon flash light
 - (d) Both (a) and (b)
5. Which of the following devices does not allow d.c. to pass through?
 - (a) resistor
 - (b) capacitor
 - (c) inductor
 - (d) all the above

[Turn Over

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6. The distance of closest approach of α -particle reaching a nucleus with momentum 'p' is r_0 .
When the α -particle travels towards the same nucleus with momentum $\frac{p}{2}$, the distance of closest approach will be :
- (a) $4r_0$ (b) $\frac{r_0}{4}$ (c) $2r_0$ (d) $\frac{r_0}{2}$
7. If β is the bandwidth, in Young's double slit experiment, the distance between the first dark band and sixth bright band is:
- (a) $5\frac{1}{2}\beta$ (b) 6β (c) 11β (d) 5β
8. According to the laws of Boolean algebra, the expression $(A+AB)$ is equal to:
- (a) B (b) $\bar{A}$ (c) A (d) AB
9. Point charges $1\mu\text{C}$ and $6\mu\text{C}$ are placed in air at a certain distance apart. The magnitude of the force on $1\mu\text{C}$ by $6\mu\text{C}$ is F_1 . The magnitude of the force on $6\mu\text{C}$ by $1\mu\text{C}$ is F_2 .
Then $F_1 : F_2$ is :
- (a) 1:1 (b) 36:1 (c) 1:6 (d) 6:1
10. High frequency waves follow:
- (a) ionospheric propagation (b) the curvature of the earth
(c) the ground wave propagation (d) the line of sight direction
11. When a hydrogen atom absorbs an energy of 10.2 eV, the change in its angular momentum is:
- (a) $4.14 \times 10^{-15} \text{ Js}$ (b) $0.525 \times 10^{-34} \text{ Js}$
(c) $1.05 \times 10^{-34} \text{ Js}$ (d) $2.1 \times 10^{-34} \text{ Js}$
12. Avalanche breakdown is primarily dependent on the phenomenon of :
- (a) doping (b) recombination
(c) collision (d) ionization
13. The alternating current in a circuit is given by the equation $i = 10\sin\left(100\pi t + \frac{\pi}{6}\right)$.
The current attains its first maximum at t is :
- (a) $\frac{1}{600} \text{ s}$ (b) $\frac{1}{50} \text{ s}$ (c) $\frac{1}{100} \text{ s}$ (d) $\frac{1}{300} \text{ s}$
14. An electric bulb is marked 220 V, 100W. When it is connected across 110 V, its power is:
- (a) 200 W (b) 173.2 W (c) 50 W (d) 25 W

15. Phosphor-bronze wire is used for suspension in a moving coil galvanometer, because it has:
- | | |
|---------------------------------|---------------------------------|
| (a) large couple per unit twist | (b) small couple per unit twist |
| (c) high conductivity | (d) high resistivity |

PART – II

Note : Answer **any six** questions. Question No. **24** is **compulsory**.

6x2=12

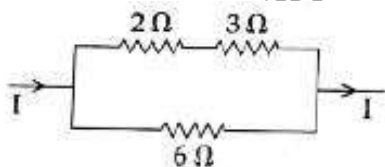
16. Define electric dipole moment. Give its unit.
17. State Ohm's Law.
18. Define Peltier Coefficient.
19. State De-Morgan's theorems.
20. Write the uses of infra-red rays.
21. What are the characteristics of laser?
22. The de-Broglie wavelength of a neutron of kinetic energy K is λ . When its kinetic energy is $4K$, what is the de-Broglie wavelength of the neutron?
23. Define curie.
24. The number of turns in the primary of an ideal transformer is 400 and that in the secondary is 2000. If the output power from the secondary at 1000 V is kW then calculate the voltage and current in the primary coil.

PART – III

Note : Answer **any six** questions. Question No. **33** is **compulsory**.

6x3=18

25. Write the properties of electric lines of forces.
- 26.



The heat developed across $6\ \Omega$ resistor per second is 50 J. Calculate the heat developed per second across $2\ \Omega$ resistor in the given electric circuit.

27. Write the special features of Magnetic Lorentz force.
28. Obtain an expression for the energy associated with an inductor.
29. Explain frequency modulation.
30. State and obtain Bragg's Law
31. Explain length contraction.

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32. Half lives of two radioactive elements are 12 hrs and 16 hrs respectively. If at any instant, the ratio of the amounts of radioactive substance is 2 : 1 , then after 2 days, What will be the ratio of the un-decayed portions?
33. In Young's double slit experiment two coherent sources of intensity ratio of 64:1, produce interference fringes. Calculate the ratio of maximum and minimum intensities.

PART – IV

Note : Answer **all** the questions.

5x5=25

34. (a) Derive an expression for electric field intensity due to an electric dipole at a point on its axial line.
- (OR)**
- (b) Obtain an expression for the magnetic induction at a point due to an infinitely long straight conductor carrying current.
35. (a) State Faraday's II Law of electrolysis. How is it verified experimentally?
- (OR)**
- (b) Explain Raman Scattering of Light.
36. (a) Discuss with theory the method of inducing emf in a coil by changing its orientation with respect to the direction of the magnetic field.
- (OR)**
- (b) Explain the working of a half wave diode rectifier.
37. (a) Explain the spectral series of hydrogen atom. (Diagram not necessary)
- (OR)**
- (b) Explain the function of AM radio transmitter with neat block diagram.
38. (a) Explain the construction and working of a Geiger-Muller Counter.
- (OR)**
- (b) Explain the working of photo emissive cell. Write any two applications of photoelectric cells.

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B**PART – III**, awg a y / **PHYSICS**

(English Version)

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[Maximum Marks : 70

- Instructions :**
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PART – I

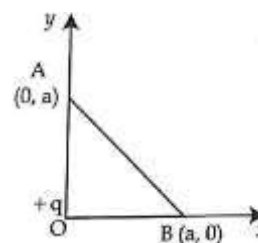
- Note :**
- Answer **all** the questions. **15x1=15**
 - Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

- The frequency range of 30 MHz to 400 GHz is used for :
 - Satellite communication
 - Ground wave propagation
 - Space wave propagation
 - Sky wave propagation
- In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is :
 - Q
 - $\frac{Q}{2}$
 - $\frac{Q}{\sqrt{3}}$
 - $\frac{Q}{\sqrt{2}}$
- Type of material which emits white light in LED :
 - GaInN
 - SiC
 - AlGaP
 - GaAsP
- The force experienced by a particle having mass m and charge q accelerated through a potential difference V when it is kept under perpendicular magnetic field $\vec{B}$ is
 - $\sqrt{\frac{2q^3BV}{m}}$
 - $\sqrt{\frac{q^3B^2V}{2m}}$
 - $\sqrt{\frac{2q^3BV}{m^3}}$
 - $\sqrt{\frac{2q^3B^2V}{m}}$
- Charging current for a capacitor is 0.2 A, find the displacement current.
 - zero
 - 0.2 A
 - 0.4 A
 - 0.1 A
- In Bohr Atom Model when the principal quantum number (n) increases the velocity of electron :
 - increases and then decreases
 - increases
 - decreases
 - remains constant
- In the given diagram a point charge +q is placed at the origin O. Work done in taking another point charge – Q from point A to point B is:

- $\frac{qQ}{4\pi\epsilon_0 a^2} \left(\frac{a}{\sqrt{2}} \right)$
- $\left[\frac{-qQ}{4\pi\epsilon_0} \frac{1}{a^2} \right] \sqrt{2}a$

(b) zero

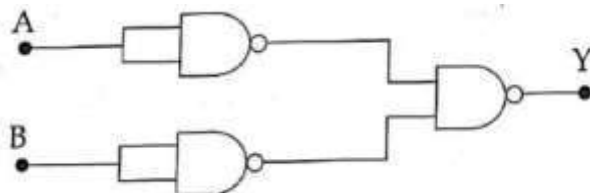
- $\left[\frac{qQ}{4\pi\epsilon_0} \frac{1}{a^2} \right] \sqrt{2}a$



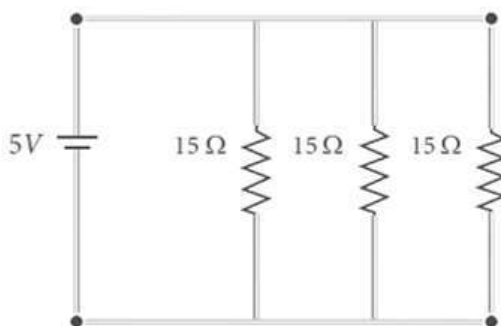
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8. The nucleus is approximately spherical in shape. Then the surface area of nucleus having mass number A varies as
- (a) $A^{\frac{5}{3}}$ (b) $A^{\frac{2}{3}}$ (c) $A^{\frac{4}{3}}$ (d) $A^{\frac{1}{3}}$
9. Two light waves from slit S_1 and S_2 on reaching points P and Q on a screen in Young's double slit experiment have a path difference zero and $\frac{\lambda}{4}$ respectively. The ratio of light intensities at P and Q will be :
- (a) 4 : 1 (b) 3 : 2 (c) $\sqrt{2} : 1$ (d) 2 : 1
10. The radius of curvature of curved surface at a thin Plano convex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be,
- (a) 20 cm (b) 5 cm (c) 10 cm (d) 15 cm
11. The given electrical network is equivalent to :



- (a) NAND gate (b) OR gate (c) NOT gate (d) Ex-OR gate
12. Magnetic field at any point at a distance R due to a long straight conductor carrying current varies as :
- (a) R^2 (b) R (c) $\frac{1}{R^2}$ (d) $\frac{1}{R}$
13. If voltage applied on a capacitor is increased from V to $2V$, choose the correct conclusion.
- (a) Both Q and C remain the same (b) Q remains the same, C is doubled
 (c) Q is doubled, C is doubled (d) C remains the same, Q is doubled
14. A light of wavelength 500 nm is incident on a sensitive plate of photoelectric work function 1.235 eV. The kinetic energy of the photo electrons emitted is :
 (Take $h = 6.6 \times 10^{-34}$ Js)
- (a) 1.16 eV (b) 0.58 eV (c) 2.48 eV (d) 1.24 eV
15. What is the current drawn out from the battery?



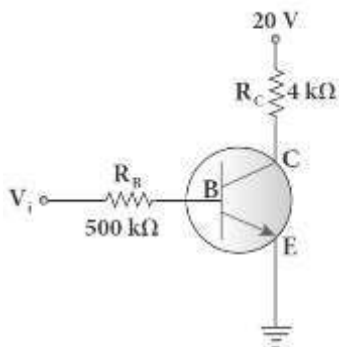
- (a) 4A (b) 1A (c) 2A (d) 3A

PART – II**Note :** Answer **any six** questions. Question No. **24** is **compulsory**.**6x2=12**

16. What do you mean by doping?
17. What are the uses of X-rays?
18. An ideal transformer has 460 and 40,000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary coil if the transformer is connected to a 230 V AC main.
19. Distinguish between Fresnel and Fraunhofer types of diffraction.
20. What is corona discharge?
21. What is skip area?
22. What are the properties of neutrino?
23. Two materials X and Y are magnetized whose intensity of magnetization are 500 Am^{-1} and 2000 Am^{-1} respectively. The magnetizing field is 1000 Am^{-1} . What is the ratio between the susceptibilities of the two material?
24. Why electron is preferred over X-ray in microscope?

PART – III**Note :** Answer **any six** questions. Question No. **33** is **compulsory**.**6x3=18**

25. Explain the conversion of galvanometer into voltmeter.
26. The resistance of a nichrome wire at 0°C is 10Ω . If its temperature coefficient of resistance is $0.004/^\circ\text{C}$, find its resistance at boiling point of water. Comment on the result.
27. What are the important inferences from the average binding energy curve?
28. In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0 \text{ V}$ and $V_{CE} = 0 \text{ V}$. What are the values of I_B , I_C , β ?



29. Derive the expression for equivalent capacitance, when capacitors are connected in parallel.
30. What are the advantages and disadvantages of AC over DC?
31. Two light sources of equal amplitudes interfere with each other. Calculate the ratio of maximum and minimum intensities.
32. Derive an expression for de-Broglie wavelength of electrons.
33. Modulation helps to reduce the antenna size in wireless communication – Explain.

[Turn Over

M2020**4****PART – IV****Note :** Answer **all** the questions.**5x5=25**

34. (a) Obtain the expression for the induced emf by changing relative orientation of the coil with the magnetic field (Graph not necessary?)

(OR)

- (b) Derive the mirror equation and the equation for lateral magnification.

35. (a) Deduce the expression for the force between two long parallel current carrying conductors.

(OR)

- (b) Write down Maxwell equations in integral form.

36. (a) Describe Davission – Germer experiment which demonstrated the wave nature of Electrons.

(OR)

- (b) (i) Derive an expression for the orbital energy of an electron in hydrogen atom using Bohr theory.
(ii) An electron in Bohr's hydrogen atom has an energy of -3.4 eV. What is the angular momentum of the electron?

37. (a) Explain the working of the transistor as an oscillator.

(OR)

- (b) Find out the phase relationship between voltage and current in a pure inductive circuit.

38. (a) State Gauss Law in electrostatics. Obtain an expression for Electric field due to an infinitely long charged wire.

(OR)

- (b) How the emf of two cells compared using potentiometer?

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**PART – III**, awg a y / **PHYSICS**

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PART – I

- Note :**
- (i) Answer **all** the questions. **15x1=15**
 - (ii) Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

1. Two identical conducting balls having positive charges q_1 and q_2 are separated by a centre to centre distance r . If they are made to touch each other and then separated to the same distance, the force between them will be :
 - (a) less than before
 - (b) same as before
 - (c) more than before
 - (d) zero
2. In an ac circuit voltage and current are given by $V = 50 \sin 50t$ Volt and $i = 100 \sin (50t) + \frac{\pi}{3}$ A. The power dissipated in the circuit will be:
 - (a) 2.5 kW
 - (b) 1.25 kW
 - (c) 5 kW
 - (d) 500 W
3. The nucleus is approximately spherical in shape. Then the volume of nucleus having mass number A varies as :
 - (a) A
 - (b) $A^{4/3}$
 - (c) $A^{1/3}$
 - (d) $A^{5/3}$
4. The wavelength λ_e of an electron and λ_p of a photon of same energy E are related by
 - (a) $\lambda_p \propto \lambda_e$
 - (b) $\lambda_p \propto \sqrt{\lambda_e}$
 - (c) $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$
 - (d) $\lambda_p \propto \lambda_e^2$
5. The resistance of a uniform wire of length l and cross-sectional area A , is R . The resistance of wire of the same material having length $2l$ and cross-sectional area $2A$ is :
 - (a) R
 - (b) $2R$
 - (c) $\frac{R}{2}$
 - (d) $\frac{R}{4}$

[Turn Over

6. Two polaroid's P_1 and P_2 are placed with their optic axes perpendicular to each other. If an un-polarized light of intensity I_0 is incident on the first polaroid P_1 then the intensity of transmitted light through the second polaroid P_2 will be :
- (a) $\frac{I_0}{2}$ (b) $\frac{I_0}{4}$ (c) 0 (d) $\frac{I_0}{8}$
7. A parallel plate capacitor stores a charge Q at a voltage V . Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
- (a) capacitance (b) Charge (c) Voltage (d) Energy density
8. The value of Bohr magneton μ_B is :
- (a) $9.27 \times 10^{-24} \text{ Am}$ (b) $9.27 \times 10^{24} \text{ Am}^{-1}$
(c) $9.27 \times 10^{24} \text{ Am}^{-2}$ (d) $9.27 \times 10^{-24} \text{ Am}^2$
9. The variation of frequency of carrier wave with respect to the amplitude of the modulating signal is called:
- (a) amplitude modulation (b) phase modulation
(c) frequency modulation (d) pulse width modulation
10. A system consists of N_0 nucleus at $t = 0$. The number of nuclei remaining after half of half-life (that is, at time $t = \frac{1}{2} T_{1/2}$)
- (a) $\frac{N_0}{2}$ (b) $\frac{N_0}{\sqrt{2}}$ (c) $\frac{N_0}{4}$ (d) $\frac{N_0}{8}$
11. When the current changes from 2 A to - 2 A in 0.05 s, an emf of 8 V is induced in a coil. The coefficient of self-induction of the coil is:
- (a) 0.2 H (b) 0.4 H (c) 0.8 H (d) 0.1 H
12. The average energy density of an electromagnetic wave is:
- (a) $\frac{1}{2} \epsilon_0 E$ (b) $\frac{1}{2} \epsilon_0 E^2$ (c) $\frac{1}{4} \epsilon_0 E^2$ (d) $\frac{1}{4} \epsilon_0 E$
13. The threshold wavelength for a metal surface whose photoelectric work function is 3.313 eV is :
- (a) 4125 Å (b) 37350 Å (c) 6000 Å (d) 2062.5 Å
14. The dopant to be added with a pure Germanium Crystal to form n type semiconductor is:
- (a) Boron (b) Phosphorus (c) Aluminium (d) Indium
15. If a beam of un-polarized light is incident on a reflecting glass surface at an angle of 57.5° , then the angle between the reflected and refracted beam will be :
- (a) 45° (b) 60° (c) 90° (d) 30°

PART – II

Note : Answer **any six** questions. Question No. **24** is **compulsory**.

6x2=12

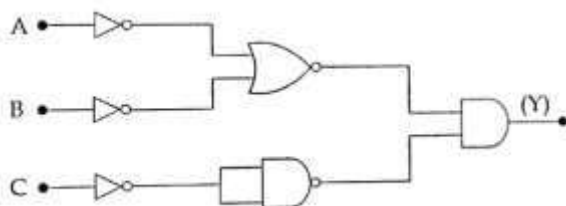
16. Define skip distance.
17. Calculate the cut-off wavelength and cut-off frequency of X-rays from an X-ray tube of accelerating potential 20,000 V.
18. State Lenz's law.
19. Potential in a given region is given as a function of distance x , $V=5(x^2+x)$ Volt. Find the electric field when $x = 1$ cm.
20. What is Photovoltaic cell?
21. What are paraxial rays and marginal rays?
22. What are the constituent particles of Neutron and Proton?
23. What are the uses of X-rays?
24. If an electric field of magnitude 570 NC^{-1} is applied in the copper wire, find the acceleration experienced by the electron.

PART – III

Note : Answer **any six** questions. Question No. **33** is **compulsory**.

6x3=18

25. Half lives of two radioactive elements A and B are 20 minutes and 40 minutes respectively. Initially the samples have equal number of nuclei. Calculate the ratio of decayed number of A and B nuclei after 80 minutes.
26. State Kirchhoff's Current and Voltage laws.
27. An $500 \mu\text{H}$, $\frac{80}{\pi^2} \text{ pF}$ capacitor and a 628Ω resistor are connected to form a series RLC circuit. Calculate the resonant frequency and Q-factor of this circuit at resonance.
28. Obtain Gauss's law of electrostatics from Coulomb's inverse square law.
29. Compare the properties of dia, para and ferromagnetic materials.
30. Fibre optic communication is gaining popularity among various transmission media. Justify.
31. What are the characteristics of photons?
32. Write the output (Y) Boolean expression for the following circuit with inputs A, B and C.



33. What is total internal reflection? Give the condition for the total internal reflection takes place.

[Turn Over

Note : Answer **all** the questions.

5x5=25

34. (a) Explain the construction and working of transformer and define its efficiency.

(OR)

- (b) Derive the equation for the angle of deviation produced by a prism and thus obtain the expression for refractive index of material of the prism.

35. (a) Obtain the condition for bridge balance in Wheatstone's bridge.

(OR)

- (b) (i) State Ampere's Circuital Law.
(ii) Find the magnetic induction due to a long straight conductor using Ampere's Circuital Law.

36. (a) Derive an expression for the radius of the orbit of electron in an atom using Bohr atom model.

(OR)

- (b) (i) Write down any six properties of electromagnetic wave.
(ii) Compute the speed of electromagnetic wave in a medium if the amplitudes of electric and magnetic fields in it are $3 \times 10^4 \text{ NC}^{-1}$ and $2 \times 10^{-4} \text{ T}$ respectively.

37. (a) Obtain Lens Maker's Formula, from that derive Lens equation.

(OR)

- (b) Describe the function of transistor as an amplifier with the neat circuit diagram. Sketch the input and output waveform.

38. (a) (i) Obtain Einstein's Photoelectric equation with necessary explanation.
(ii) What will happen to the stopping potential in the following cases when;
(A) Work function of the metal is increased.
(B) Intensity of incident ray is increased.

(OR)

- (b) Explain in detail the effect of introducing a dielectric medium between the plates of a parallel plate capacitor, when the capacitor is disconnected from the battery.

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B

PART – III

, awgpay / PHYSICS

(English Version)

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[Maximum Marks : 70

- Instructions :**
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PART – I

- Note :**
- (i) Answer **all** the questions. **15x1=15**
 - (ii) Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

1. In an oscillating LC circuit, the maximum charge on the capacitor is Q. the charge on the capacitor when the energy is stored equally between the electric and magnetic field is:
 - (a) $\frac{Q}{\sqrt{2}}$
 - (b) $\frac{Q}{2}$
 - (c) Q
 - (d) $\frac{Q}{\sqrt{3}}$
2. In a Young's double slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to:
 - (a) $\sqrt{2} D$
 - (b) 2D
 - (c) $\frac{D}{\sqrt{2}}$
 - (d) $\frac{D}{2}$
3. Which charge configuration produces a uniform electric field?
 - (a) Uniformly charged infinite plate
 - (b) point charge
 - (c) Uniformly charged spherical shell
 - (d) Uniformly charged infinite line
4. The ratio of magnetic length and geometrical length is :
 - (a) 0.833
 - (b) 0.633
 - (c) 0.933
 - (d) 0.733
5. The internal resistance of a 2.1 V cell which gives a current of 0.2A through a resistance of 10 Ω is :
 - (a) 0.8 Ω
 - (b) 0.2 Ω
 - (c) 1.0 Ω
 - (d) 0.5 Ω
6. If the nuclear radius of ^{27}Al is 3.6 fermi, the approximate nuclear radius of ^{64}Cu in fermi is:
 - (a) 4.8
 - (b) 2.4
 - (c) 3.6
 - (d) 1.2

[Turn Over

7. If the velocity and wavelength of light in air is V_a and λ_a and that in water is V_w and λ_w , then the refractive index of water is,
- (a) $\frac{\lambda_w}{\lambda_a}$ (b) $\frac{V_w}{V_a}$ (c) $\frac{V_a \lambda_a}{V_w \lambda_w}$ (d) $\frac{V_a}{V_w}$
8. The unit of electric flux is :
- (a) Nm^{-1}C^2 (b) $\text{Nm}^{-2}\text{C}^{-1}$ (c) Nm^2C^{-1} (d) N^2mC^{-1}
9. For a healthy eye, the distance of the near point is _____
- (a) 30 cm (b) 20 cm (c) 35 cm (d) 25 cm
10. The blueprint for making ultra-durable synthetic material is mimicked from:
- (a) Parrot fist (b) Lotus leaf
(c) Peacock feather (d) Morpho butterfly
11. Emission of electrons by the absorption of heat energy is called _____
- (a) Thermionic (b) Photoelectric
(c) Secondary (d) Field
12. The Zener diode is primarily used as :
- (a) Oscillator (b) Rectifier (c) Voltage regulator (d) Amplifier
13. Which of the following is false for electromagnetic waves?
- (a) longitudinal (b) transverse
(c) produced by accelerating charges (d) non-mechanical waves
14. The force experienced by a particle having mass m and charge q accelerated through a potential difference V when it is kept under perpendicular magnetic field $\vec{B}$.
- (a) $\sqrt{\frac{2q^3B^2V}{m}}$ (b) $\sqrt{\frac{2q^3BV}{m}}$ (c) $\sqrt{\frac{2q^3BV}{m^3}}$ (d) $\sqrt{\frac{q^3B^2V}{2m}}$
15. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6 A, then that in the secondary coil is :
- (a) 12 A (b) 2 A (c) 1 A (d) 18 A

PART – II

Note : Answer **any six** questions. Question No. **24** is **compulsory**.

6x2=12

16. Mention the ways of producing induced emf.
17. Find the Polarizing angle for glass of refractive index 1.5
18. What is Peltier effect?
19. Define “Electrostatic Potential”.
20. How will you define threshold frequency?
21. State Ampere’s Circuital Law.
22. Why does sky appear blue?
23. Give two uses of IR radiation.
24. Dielectric strength of air is $4 \times 10^6 \text{ Vm}^{-1}$. Suppose the radius of a hollow sphere in the Van de Graaff generator is $R = 0.4 \text{ m}$, calculate the maximum potential difference created by this Van de Graaff generator.

PART – III

Note : Answer **any six** questions. Question No. **33** is **compulsory**.

6x3=18

25. State Kirchhoff’s current and voltage rule.
26. What are critical angle and total internal reflection?
27. List out the characteristics of Photons.
28. Obtain the expression for energy stored in the parallel plate capacitor.
29. Mention the differences between interference and diffraction.
30. The repulsive force between two magnetic poles in air is $9 \times 10^{-3} \text{ N}$. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.
31. Draw the circuit diagram of a full wave rectifier and draw its input and output waveforms.
32. Mention the various energy losses in a transformer.
33. ${}^{92}\text{U}^{235}$ nucleus emits 2α particles, 3β particles and 2γ particles. What is the resulting atomic number and mass number?

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PART – IV

Note : Answer **all** the questions.

5x5=25

34. (a) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current.

(OR)

- (b) Obtain the law of radioactivity.

35. (a) Calculate the electric field due to a dipole on its axial line.

(OR)

- (b) What is Frequency Modulation? List out the advantages and limitations of frequency modulation.

36. (a) (i) Derive an expression for de-Broglie wavelength of electrons.
(ii) Calculate the momentum of an electron with kinetic energy 2 eV.

(OR)

- (b) Write down Maxwell equations in integral form.

37. (a) Explain about Astronomical telescope and obtain the equation for the magnification.

(OR)

- (b) (i) Explain the equivalent resistance of a series resistor network.
(ii) A copper wire of cross-sectional area 0.5 mm^2 carries a current of 0.2 A. If the free electron density of copper is $8.4 \times 10^{28} \text{ m}^{-3}$ then compute the drift velocity of free electrons.

38. (a) Obtain Lens maker's formula.

(OR)

- (b) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

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**PART – III**, awgpa y / **PHYSICS**

(English Version)

Time Allowed : 3.00 Hours]

[Maximum Marks : 70

- Instructions :**
- (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.
 - (2) Use **Blue** or **Black** ink to write and underline and pencil to draw diagrams.

PART – I

- Note :**
- (i) Answer **all** the questions. **15x1=15**
 - (ii) Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

1. Which one of the following is the natural nanomaterial?
 - (a) Grain of sand
 - (b) Peacock feather
 - (c) Skin of the whale
 - (d) Peacock beak
2. In an electron microscope, the electrons are accelerated by a voltage of 14kV. If the voltage is charged to 224 kV, the de-Broglie wavelength associated with the electrons would:
 - (a) decrease by 4 times
 - (b) increase by 2 times
 - (c) increase by 4 times
 - (d) decrease by 2 times
3. The variation of frequency of carrier wave with respect to the instantaneous amplitude of the modulating signal is called:
 - (a) Phase modulation
 - (b) Amplitude modulation
 - (c) Pulse width modulation
 - (d) Frequency modulation
4. Q factor is equal to _____.
 - (a) $\frac{\omega_r L}{R}$
 - (b) $\frac{1}{R} \sqrt{\frac{L}{C}}$
 - (c) $\frac{X_L}{R}$
 - (d) All the above

[Turn Over

M2022

5. The metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is:
- (a) 1×10^{-2} C (b) 3×10^{-2} C
(c) 2×10^{-2} C (d) 4×10^{-2} C
6. Which of the following is an electromagnetic wave?
- (a) β – rays (b) γ – rays
(c) α – rays (d) All of the above
7. An air bubble in glass slab of refractive index 1.5 (near normal incidence) is 5 cm deep when viewed from one surface and 3 cm deep when viewed from the opposite face. The thickness of the slab is :
- (a) 12 cm (b) 8 cm
(c) 16 cm (d) 10 cm
8. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60 W bulb for use in India is R, the resistance of a 60 W bulb for use in USA will be :
- (a) $R/4$ (b) R
(c) $R/2$ (d) $2R$
9. A wire of length l carries a current I along the Y direction and magnetic field is given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})T$. The magnitude of Lorentz force acting on the wire is
- (a) $\sqrt{2}\beta Il$ (b) $\sqrt{\frac{2}{\sqrt{3}}}\beta Il$
(c) $\sqrt{\frac{1}{\sqrt{2}}}\beta Il$ (d) $\sqrt{\frac{1}{\sqrt{3}}}\beta Il$
10. Emission of electrons by the absorption of heat energy is called _____ emission.
- (a) Thermionic (b) Photo electric
(c) Secondary (d) Field
11. If a current of 7.5 A is maintained in a wire for 45 seconds then the charge flowing through the wire is :
- (a) 6 C (b) 365.5 C
(c) 3 C (d) 337.5 C
12. The charge of cathode ray is :
- (a) neutral (b) positive
(c) not defined (d) negative

13. A step-down transformer reduces the supply voltage from 220 V to 11 V and increases the current from 6 A to 100 A. Then its efficiency is :
- (a) 0.12 (b) 1.2
(c) 0.9 (d) 0.83
14. The electric potential between a proton and an electron is given by $V = V_0 \ln \left(\frac{r}{r_0} \right)$, where r_0 is a constant. Assume that Bohr atom model is applicable to potential, then variation of radius of n^{th} orbit r_n with the principal quantum number n is
- (a) $r_n \propto \frac{1}{n^2}$ (b) $r_n \propto \frac{1}{n}$
(c) $r_n \propto n^2$ (d) $r_n \propto n$
15. Transverse nature of light is shown in :
- (a) scattering (b) interference
(c) polarization (d) diffraction

PART – II

Note : Answer **any six** questions. Question No. **24** is **compulsory**.

6x2=12

16. What is corona discharge?
17. How will you increase the current sensitivity of a galvanometer?
18. Define work function of a metal. Mention its unit.
19. Calculate the radius of $^{197}_{79}\text{Au}$ nucleus.
20. State Fleming's right hand rule.
21. What do you mean by Doping?
22. What is displacement current?
23. Define electrical resistivity.
24. The angle of minimum deviation for the equilateral prism is 40° . Find the refractive index of the material of the prism.

PART – III

Note : Answer **any six** questions. Question No. **33** is **compulsory**.

6x3=18

25. Derive the relation between f and R for a spherical mirror.
26. Obtain a relation between current and drift velocity.
27. List out the laws of photo electric effect.
28. Draw the circuit diagram of NPN transistor in Common Emitter Configuration.
29. Give the uses of Polaroids.
30. Derive the expression for resultant capacitance, when capacitors are connected in series.

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31. Find the:
- Angular momentum
 - Velocity of the electron revolving in the 5th orbit of hydrogen atom.
($h = 6.6 \times 10^{-34}$ Js; $m = 9.1 \times 10^{-31}$ kg)
32. List out salient features of magnetic Lorentz force.
33. Find the impedance of a series RLC circuit, if the inductive reactance, capacitive reactance and resistance are 184Ω , 144Ω , and 30Ω respectively. Also calculate the phase angle between voltage and current.

PART – IV**Note :** Answer **all** the questions.**5x5=25**

34. (a) Explain the construction and working of full wave rectifier.
(OR)
(b) Explain the construction and working of transformer.
35. (a) Derive an expression for electrostatic potential due to an electric dipole.
(OR)
(b) Obtain the equation for bandwidth in Young's Double Slit Experiment.
36. (a) Using Biot-Savart Law deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current.
(OR)
(b) Discuss the spectral series of hydrogen atom.
37. (a) (i) How do you we obtain characteristic X-ray spectra?
(ii) Calculate the cut-off wavelength and cut-off frequency of X-rays from an X-ray tube of accelerating potential 20,000 V.
(OR)
(b) What is spectrum? Explain the types of emission spectrum.
38. (a) Obtain Lens maker's formula.
(OR)
(b) Explain the determination of the internal resistance of cell using voltmeter.

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**PART – III**

, awgpay / **PHYSICS**
(English Version)

Time Allowed : 3.00 Hours]

[Maximum Marks : 70

- Instructions :**
- (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.
 - (2) Use **Blue** or **Black** ink to write and underline and pencil to draw diagrams.

PART – I

- Note :**
- (i) Answer **all** the questions. **15x1=15**
 - (ii) Choose the most appropriate answer from the given **four** alternatives and write the option code and the corresponding answer.

1. The ratio between the radius of first three orbits of hydrogen atom is:
(a) 1:2:3 (b) 1:2:2 (c) 1:4:9 (d) 1:3: 5
2. Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are:
(a) $5I$ and I (b) $5I$ and $3I$ (c) $9I$ and I (d) $9I$ and $3I$
3. A wire connected to a power supply of 230 V has power dissipation P_1 . Suppose the wire is cut into two equal pieces and connected parallel to the same power supply. P_2 In this case power dissipation is P_2 The ratio $\frac{P_2}{P_1}$ is :
(a) 1 (b) 2 (c) 3 (d) 4
4. Stars twinkle due to:
(a) Reflection (b) Total internal reflection
(c) Refraction (d) Polarisation
5. The instantaneous values of alternating current and voltage in a circuit are
 $i = \frac{1}{\sqrt{2}} \sin(100\pi t)$ A and $v = \frac{1}{\sqrt{2}} \sin\left(100\pi t + \frac{\pi}{3}\right)$ V.
The average power in watts consumed in the circuit is:
(a) $\frac{1}{4}$ (b) $\frac{\sqrt{3}}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
6. If the mean wavelength of light from Sun is taken as 550 nm and its mean power as 3.8×10^{26} W then, the average number of photons received by the human eye per second from Sunlight is of the order of:
(a) 10^{45} (b) 10^{42} (c) 10^{54} (d) 10^{51}

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7. An electric dipole is placed at an alignment angle of 30° with an electric field of $2 \times 10^5 \text{ NC}^{-1}$. It experiences a torque equal to 8 Nm. The charge on the dipole if the dipole length is 1 cm is
(a) 4 mC (b) 8 mC (c) 5 mC (d) 7 mC
8. Fraunhofer lines are an example of spectrum.
(a) line emission (b) line absorption
(c) band emission (d) band absorption
9. The mass of a ${}^7_3\text{Li}$ nucleus is 0.042 u less than the sum of the masses of all its 3 nucleons. The average binding energy per nucleon ${}^7_3\text{Li}$ nucleus is nearly :
(a) 46 MeV (b) 5.6 MeV (c) 3.9 MeV (d) 23 MeV
10. The temperature co-efficient of resistance of a wire is $0.00125 \text{ per } ^\circ\text{C}$. At 20°C , its resistance is 1Ω . The resistance of the wire will be 2Ω at
(a) 800°C (b) 700°C (c) 850°C (d) 820°C
11. The particle size of ZnO material is 30 nm. Based on the dimension it is classified as :
(a) Bulk material (b) Nanomaterial (c) Soft material (d) Magnetic material
12. The value of L, C and R of an AC circuit are 1 H, 9 F and 3Ω respectively. The quality factor for this circuit is :
(a) 1 (b) 9 (c) $\frac{1}{9}$ (d) $\frac{1}{3}$
13. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
(a) 1.0 Am^2 (b) 1.2 Am^2 (c) 0.5 Am^2 (d) 0.8 Am^2
14. Two radiations with photon energies 0.9 eV and 3.3 eV respectively are falling on a metallic surface successively. If the work function of the metal is 0.6 eV, then the ratio of maximum speeds of emitted electrons in the two cases will be
(a) 1:4 (b) 1:3 (c) 1: 1 (d) 19
15. If a positive half-wave rectified voltage is fed to a load resistor, for which part of a cycle of the input signal there will be current flow through the load?
(a) $0^\circ - 90^\circ$ (b) $90^\circ - 180^\circ$ (c) $0 - 180^\circ$ (d) $0^\circ - 360^\circ$

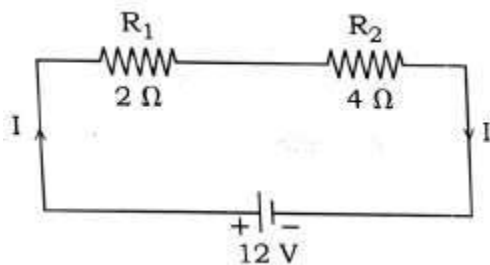
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PART – II

J2022

Note : Answer **any six** questions. Question No. **24** is **compulsory**.

6x2=12

16. Mention the ways of producing induced emf.
17. Define stopping potential.
18. Give two uses of UV radiation.
19. Pure water has refractive index 1.33. What is the speed of light through it ?
20. Define ampere in terms of force.
21. What is rectification?
22. State Gauss law.
23. Define atomic mass unit.
24. Calculate the equivalent resistance for the circuit which is connected to 12 V battery and also find the potential difference across $2\ \Omega$ and $4\ \Omega$ resistors in the circuit.



PART – III

Note : Answer **any six** questions. Question No. **33** is **compulsory**.

6x3=18

25. Obtain an expression for energy stored in the parallel plate capacitor.
26. An electron moving perpendicular to a uniform magnetic field 0.500 T undergoes circular motion of radius 2.50 mm. What is the speed of electron?
27. Give the construction and working of a photo emissive cell.
28. Mention the differences between interference and diffraction.
29. What is Zener diode? Mention any two uses of Zener diode.
30. What is Seebeck effect? State the applications of Seebeck effect.
31. What are the properties of Cathode rays?
32. AC is advantageous than DC. Explain.
33. Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5. What is the speed of light in the glass slab and what is the time taken by the light to travel through the glass slab.

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J2022**4****PART – IV****Note :** Answer **all** the questions.**5x5=25**

34. (a) Obtain the condition for bridge balance in Wheatstone's bridge.
(OR)
(b) (i) What is half-life and mean life of a radioactive nucleus?
(ii) Calculate the number of nuclei of carbon-14 un-decayed after 22,920 years if the initial number of carbon-14 atoms is 10,000. The half –life of carbon-14 is 5730 years.
35. (a) Describe the Fizeau's method to determine the speed of light.
(OR)
(b) (i) Write down the properties of electromagnetic waves.
(ii) The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.
36. (a) Explain in detail the construction and working of a Van de Graaff generator.
(OR)
(b) Explain about Compound Microscope and obtain the equation for the Magnification.
37. (a) Show that the mutual inductance between a pair of coils is same ($M_{12} = M_{21}$).
(OR)
(b) State and prove De Morgan's first and second theorem.
38. (a) (i) Obtain Einstein's photoelectric equation with necessary explanation.
(ii) List out the characteristics of photons.
(OR)
(b) Derive the expression for the force on a current carrying conductor in a magnetic field.

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